

EXPOREPORT

“ExpoCiencia 2026 Jaime Lora Jaramillo”

"STEM+ for the Caribbean: classroom-born solutions that protect the coast"

Title	The Cañaguate Tree (<i>Handroanthus billbergii</i>) as a Native Species for Soil Restoration Through the Design of Seed Bombs
Team photo	
Expositores	Samuel Rodríguez Goez Kamila Mendoza Tom Lucía Muñoz Madrid Elías Espinosa Ochoa Antonella Tarriba González
Grado	2ºA
Question	How can Seed Bombs with Cañaguate seeds help restore degraded areas and promote reforestation?
General Objective	To demonstrate how Seed Bombs with Cañaguate seeds can help restore degraded areas and support reforestation.
Specific Objectives	1. Investigate the Cañaguate Tree: Research the biological traits of the Cañaguate tree and its ability to adapt to and protect dry soils in the Caribbean region. 2. Design the "Life Bombs": Test different proportions of clay, fertile soil, and organic compost to find the strongest mixture that protects the seed while retaining moisture. 3. Evaluate and Measure the Prototype: Measure the weight, size, and drying time of the spheres to evaluate their impact resistance and estimated germination rate.

	4. Promote Interactive Reforestation: Map target areas around the school to launch the seed bombs and raise community awareness through the "Guardians of the Cañaguate" activity.
Materials	• Cañaguate seeds. • Natural clay. • Fertile soil. • Organic compost. • Water. • Mixing containers. • Protective gloves. • Measuring ruler. • Digital scale. • Measuring tape. • Plastic cups for testing. • Labels. • Camera or smartphone. • Computer. • Internet access. • Cardboard for charts. • Markers.
Procedure	S: (Science) Biological research of Cañaguate (<i>Handroanthus billbergii</i>): It is a native deciduous tree of the Caribbean region, recognized for its impressive intense yellow flowering during the dry season. It has high resistance to drought, deep root systems that prevent erosion, and a great capacity to adapt to degraded or clayey soils typical of arid and semi-arid areas of the northern Colombian coast. • Research the biological characteristics of the Cañaguate tree. • Study its life cycle and reproductive process. • Investigate its ecological importance in restoring degraded soils. • Observe different soil conditions where the species could be planted. • Formulate the project's scientific hypothesis. • Compare scientific information with field observations. T: (Technology) • Search for scientific information using reliable digital sources. • Create a digital map of the school showing where the Seed Bombs will be placed. • Document plant growth using photographs and videos. • Generate a QR code that links to the project's progress. • Organize all collected information in digital files for future analysis. E: (Engineering)

Design and manufacturing process of Life Bombs (Engineering Methodology): An iterative design cycle structured into four fundamental phases was applied to ensure the effectiveness of the reforestation spheres:

1. Selection and sieving of raw materials: Natural clay and fertile soil were sieved to remove impurities and large lumps, ensuring a homogeneous mixture that acts as a protective matrix.
 2. Formulation and mixture dosage: Prototypes were designed with different proportions of clay (50%), fertile soil (35%), and organic compost (15%) to evaluate structural cohesion and moisture retention capacity.
 3. Modeling and strategic encapsulation: Cañaguate seeds were placed in the central core of each sphere (3.5 to 4.5 cm in diameter), physically protecting them from direct solar radiation and predation by insects or birds before the rainy season.
 4. Curing, controlled drying, and impact testing: The units were subjected to a drying process in the shade for 48 to 72 hours to prevent cracking, followed by mechanical resistance tests via launch simulation to verify that the structure supports impact without premature fracturing.
- Prepare different mixtures of clay, fertile soil, and organic compost.
 - Design and build Seed Bombs containing Cañaguate seeds.
 - Allow the Seed Bombs to dry under controlled conditions.
 - Perform impact resistance tests on each design.
 - Compare the performance of each mixture.
 - Select the mixture that best protects the seed while maintaining suitable conditions for germination.

M: (Mathematics)

Metric data recording and tabulation: A quantitative recording system was incorporated to evaluate the consistency of the produced units, including the recording of approximate values per unit:

- Measure the weight of each Seed Bomb.
- Record the drying time.
- Count the number of seeds used.
- Register the germination percentage.
- Measure the height of seedlings during the observation period.
- Organize the collected data into tables.
- Create graphs comparing the results of each experimental design.
- Analyze averages and percentages to determine the most effective Seed Bomb mixture.

	Component / Variable	Approximate Value per Unit	Description / Function
	Total weight (wet)	45 g – 60 g	Initial weight of the freshly molded sphere.
	Total weight (dry)	30 g – 40 g	Final weight once water has evaporated.
	Sphere diameter	3.5 cm – 4.5 cm	Optimal size for launching and protection.
	Cañaguate seeds	2 – 3 seeds	Recommended quantity per unit.
	Natural clay	50% (approx. 20 g)	Firm structure against erosion.
	Fertile soil	35% (approx. 14 g)	Base substrate for root development.
	Fertile soil	35% (approx. 14 g)	Base substrate for root development.
	Organic compost	15% (approx. 6 g)	Nutrients for initial growth.
	Added water	10 ml – 15 ml	Moisture for moldable dough.
	Drying time	48 hours – 72 hours	Homogeneous drying in the shade.
	Estimated germination	75% – 85%	Expected rate under adequate conditions.
+: (plus) Students will build a three-dimensional model of the Cañaguate tree using recycled materials. They will also design environmental awareness posters explaining the ecological importance of native trees and encouraging the protection and restoration of ecosystems in the Colombian Caribbean region.			
Development and Evidence:			



Image 1, soil before life bombs

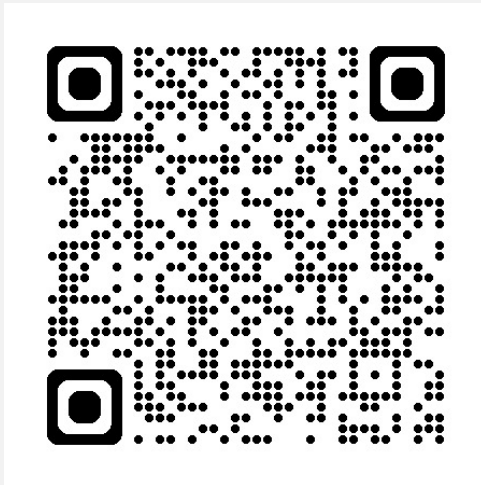


Image 2, QR code that leads to the project web page.



Image 3, students developing the life bombs.



Image 4, results of the life bombs.

ExpoPlay:

"Guardians of the Cañaguate"

Visitors will participate in an interactive environmental challenge. Guided by the students, they will prepare a small Seed Bomb using clay, soil, compost, and a Cañaguate seed.

After completing the Seed Bomb, participants will throw it onto a large map representing different areas of the school. If the Seed Bomb lands on one of the marked degraded zones, they will earn the honorary title of "Guardian of the Cañaguate."

Throughout the activity, students will explain the ecological importance of the Cañaguate tree, the scientific principles behind Seed Bomb technology, and how this simple engineering solution can support ecosystem restoration and biodiversity conservation.

**Results/
Conclusions**

The research demonstrated that the Cañaguate tree is an important native species for the Colombian Caribbean because of its ability to thrive in dry climates, protect the soil, improve environmental conditions, and support local biodiversity.

The project showed that biodegradable Seed Bombs are a practical, low-cost, and environmentally friendly method for distributing seeds in areas where traditional planting methods are difficult or impossible to apply.

Testing different mixtures of clay, fertile soil, and organic compost allowed the team to identify the combination that provided the best balance between impact resistance and moisture retention, increasing the likelihood of successful seed germination.

Data collected through measurements, observations, and comparative analysis demonstrated the importance of applying the scientific method when designing environmental solutions. Recording variables such as Seed Bomb weight, drying time, and germination rate helped students evaluate the effectiveness of each prototype.

The project successfully achieved its main objective by integrating Science, Technology, Engineering, and Mathematics into a practical solution for ecological restoration. Students also developed research skills, teamwork, critical thinking, problem-solving abilities, and environmental responsibility.

Future work should include monitoring seed growth over several months, testing Seed Bombs under different environmental conditions, and expanding the project to degraded areas throughout Cartagena to evaluate its long-term ecological impact.

The project demonstrates that small, community-based actions supported by STEM education can contribute to protecting native ecosystems and promoting sustainable environmental conservation.

